

Trajectory Generation using Profile-Specific Transition Matrices

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Abstract

Synthetic mobility data are needed where observed trajectories are costly, sparse, noisy, or privacy-sensitive. This paper explores whether large language models (LLMs) can be used not only to generate trajectories, but also to estimate profile-specific transition structures across urban buildings. We compare agents with and without a car under two autonomy regimes: a supervised regime constrained by a gravity-based transition matrix and an autonomous regime guided mainly by building attributes and prompt context. Results from 4,000 generated trajectories in Jerusalem show that profile and autonomy affect trajectory length, distance, land-use composition, and transition-graph connectivity. Supervision improves spatial coherence but also creates constraint-related errors, especially around returning home.

Keywords: LLM, trajectory generation, transition matrix

1. Introduction

Fine-grained mobility data are critically required across multiple fields for example, transportation planning, emergency management, and spatial epidemiology as their richness provides invaluable information on the behavior and the performance of urban systems. However, simultaneously multiple obstacles exist to using them such as cost, noisiness, and the protection of privacy. These constraints have made synthetic mobility generation, i.e. developing procedures for creating artificial trajectories that preserve real-world behavioral properties, an increasingly important research direction (Luca et al., 2021; Chen et al., 2025).

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The literature offers a number of approaches to trajectory generation. First, there are methods imposing interpretable structures on mobility, e.g. rule-based activity models and (hidden) Markov models (Jiang et al., 2016; Chen et al., 2025). Their appeal lies in the possibility of inspecting and comparing model parameters, but it is also constrained by the heavy reliance on behavioral assumptions. Learning-based trajectory generators circumvent this issue by fitting complex spatiotemporal distributions from data (Luca et al., 2021; Chen et al., 2025). Large language models (LLMs) have recently entered this literature, offering opportunities to enrich trajectory generation through complex semantics such as temporal context, historical stays, POI categories, traveler profiles, and natural-language place descriptions (Li et al., 2024; Hsu et al., 2024; Gong et al., 2024; Gao and Wu, 2026).

One important contextual element is the transition structure that governs mobility. Recent work (Lin et al., 2026; Grinberger et al., in press) shows that transition probabilities between building pairs are highly impactful and can encode socio-demographic and spatial semantics. However, this is rarely addressed in LLM-based trajectory generation. Another gap lies in the variety of autonomy levels granted to the model, from narrow destination choices to open-ended generation of complete routines a non-trivial modeling decision without benchmarking. Against this background, we position LLMs as a means for estimating semantic, profile-specific transition structures, asking how transition matrices change with behavioral context (socio-demographic profile) and LLM autonomy. We contribute by extending LLM use to transition dynamics and by formally testing this under-considered modeling decision.

2. Conceptual Approach

Individual real-life trajectories emerge through a series of contextual decisions. Generally speaking, the trajectory is conditioned on two types of contexts. The first is spatial, i.e. the distribution of locations within the activity space and the second behavioral, e.g. the socio-economic characteristics of the individual, movement history, and type of activity. Aggregative flows across multiple trajectories are constructed in relation to both types of contexts and hence transition matrices, as a formal representation of macro flows, encode them. For example, in a simple gravity model the spatial context is captured by origin and destination locations and concepts like economies of scale and friction of distance are encoded in the computation of probabilities. On this basis, we can reconceptualize a transition matrix as a graph in which nodes represent the spatial context and edges (and their weights, i.e. transition probabilities) denote the behavioral setting.

LLM reasoning represents a more complex and semantically-sensitive behavioral context. This can be conceived as a non-linear function of prompts and their contents, as an LLM is fundamentally a neural network that takes textual input as a sequence of tokens. The level of autonomy, while a technical rather than a behavioral definition, can certainly be considered part of the behavioral context as it uniquely frames the behavioral process. When asked to generate a trajectory or choose the next destination, we consider the sequential token generation of LLM as a sampling process that focuses on a regime of the hypothesized internal spatial distribution conditioned on the spatial context and the given prompts. When sampled enough times, it can reconstruct the conditional spatial distribution.

3. Methodology

Our approach hinges on formulating generic textual prompts, feeding them into GPT 5.4 – chosen for its reasoning capabilities – and extracting its output for downstream analysis. The prompt identifies the agent in terms of employment, age, and car ownership, and instructs the LLM to generate a trajectory of arbitrary length starting and ending at home, without revisiting any buildings (see Appendix A.1). Spatial context is established by feeding in numerical building information, including building ID, coordinates, height, floorspace and land use, in tabular format; each stop requires output of timestamp, building ID, and behavioral reasoning. The spatial context used here relates to 1,014 buildings in downtown Jerusalem, Israel, an area characterized by mixed uses, two main thoroughfares (Jaffa St. and Agripas St.) and two main activity anchors – the Mahne Yehuda market and the city center (see Appendix A.2).

Implementation scenarios differ by autonomy level and mobility profiles. Autonomy is imposed by introducing different lists of potential destination buildings per stop: the autonomous mode generates the trajectory in one shot using the full spatial context, while the supervised mode requires multiple iterations in a conversation, each choosing one stop from a list derived from a pre-computed simple gravity model (top 10% most probable transitions per building, unweighted; see Appendix A.2). Mobility profiles relate to car ownership, differentiating between agents who “have **a** car” to those who “have **no** car” (see Appendix A.1). We define 1,000 agents (age randomly 18–59, all employed) with home drawn from residential and mixed-use buildings. Each agent generates one trajectory per implementation scenario (2 autonomy regimes x 2 mobility profiles), yielding 4,000 trajectories, 1,000 per scenario.

Two limitations bear noting. The first is that utilizing a larger spatial context risks exceeding the model’s token limit, even for less token-intensive supervised prompts. Second, under supervised mode token cost compounds as the number of stops increases since all previous iterations are retained in context. These motivate constraining the implementation to the small study area used here.

4. Validation and Analysis

4.1. Reasoning, Hallucinations, and Violations

The paper’s main focus is the distribution shift across autonomy regimes and profiles; we report on the reasoning process and violations in anecdotal terms only. The requirement to return home is the primary driver of issues: small residential buildings have low in-degree values, leading the LLM to struggle with sampling the exact building ID, at times generating hallucinations, an outcome seen almost exclusively under the supervised regime and essentially absent under the autonomous one. Of the 6,637 trajectory segments under supervised mode, 81.4% (5,401) correctly include the next stop’s building ID in the reasoning. Of the 2,000 supervised trajectories, 225 violated transition dynamics: 134 did not reach home by the 11th iteration (and hence were artificially terminated) and 91 violated the final transition by outputting “home” while ignoring the adjacency list. Inspection of 38 intermediate-stop violations reveals two main causes: conflating building lists from previous steps with the current adjacency list, and numerical

imprecision in token embeddings, where the model reasons about a strong candidate but outputs a different building ID.

4.2. Distribution Shifts

We excluded from the analysis agents whose trajectories in at least one scenario included violations and hallucinations, thereby ending up with a set of 833 trajectories per scenario. These cover 585 (57.7%, supervised w/o car), 580 (57.2%, supervised w/ car), 615 (60.7%, autonomous w/ car), 622 (61.3%, autonomous w/o car) of 1,014 buildings, corresponding to four implementations. These results are not ideal, yet building coverage could be improved by, for example, increasing the number of agents or diversifying the set of trajectory origins/destinations.

Inspecting the lengths, i.e. the number of visited buildings, per implementation scenario (Figure 1), it is clear that autonomy matters, as the resulting distributions take on different shapes: the autonomous version produces a quasi-normal distribution (perhaps indicating some form of a distributional assumption) while the results for the supervised version follow a trend resembling a log-normal distribution. However, both show similarities in terms of patterns of differences between profiles, with the car-based one showing a flatter curve covering a wider range (in the case of the autonomous model, it strangely also includes shorter trajectories than the no car scenario) while differences in where the curve peaks are minimal. Notice that this does not relate to the total distance covered by trajectories, but only to the number of buildings visited. Given this, the results for the supervised implementation cases seem to be more plausible.

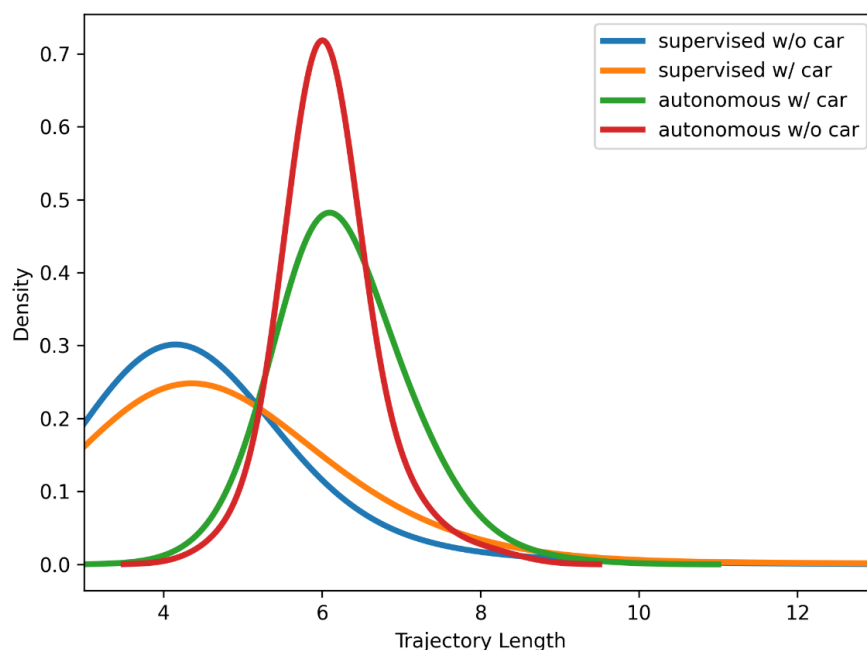


Figure 1. Distribution of trajectory lengths (number of buildings visited per trajectory) by implementation scenario.

Observing travel distance per stop (Figure 2) highlights an interesting pattern with the supervised mode showing little differences while under the autonomous regime, travelling by car leads to larger travel distances. This may point to the autonomous mode producing more plausible results,

yet the scale of the study area, where most distances are walkable, suggests that there is room for a more careful interpretation as such differences are not deterministic for such small areas. Another questionable pattern is embodied in the high frequencies of visiting land uses other than residential and commercial. Especially notable is the dominance of mixed use buildings for the second stop under the w/ car profile or the relatively high frequency of visiting elderly shelters, which are almost never visited under the supervised mode.

However, the autonomous regime also displays some meaningful patterns. First, when floorspace volumes per stop are inspected (Figure 2), it replicates the tendency observed under the supervised mode to visit larger commercial buildings early on under all implementations, suggesting that the LLM internalizes gravity model logics even when not introduced with the transition matrix. Also, under the autonomous regime, trajectories based on car usage show more diversity in land use mix across stops in relation to the no car profile. This is perhaps due to knowledge regarding access to cars increasing accessibility and hence the diversity of opportunities¹. Coupling this with the tendency to visit multiple residential locations under the supervised mode which probably emerges due to the pressure of returning home being exaggerated by the constraints of the gravity transition matrix, it becomes hard to claim that the supervised mode outperforms the autonomous approach. In any case, it is clear that distributional shifts are more meaningful under the autonomous regime.

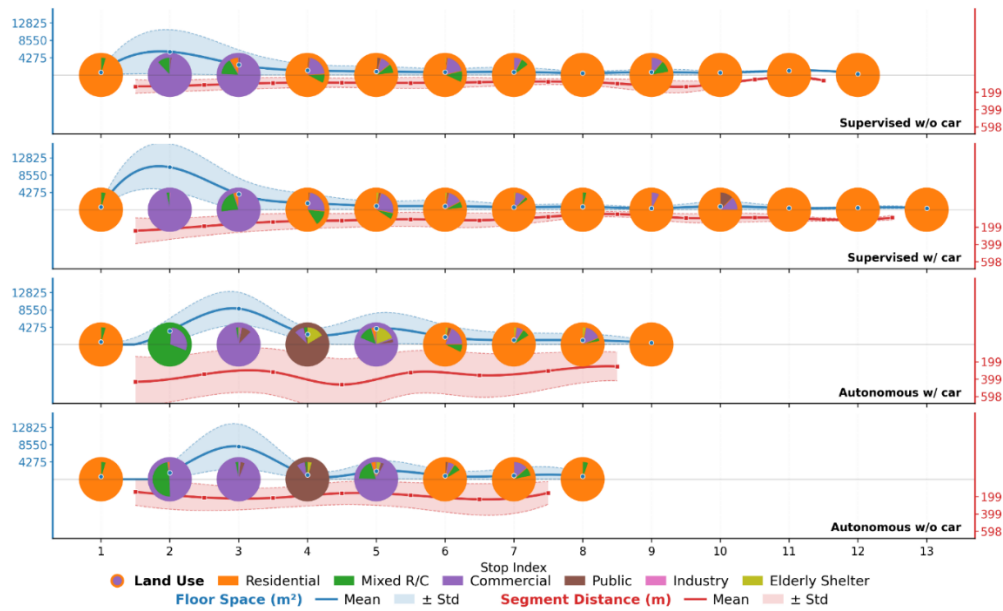


Figure 2. The mean floorspace and land use ratio of each stop together with the mean segment distance between the stops. The fill color of each circle indicates the major land use of visited buildings at each stop. Notice that lower distances appear higher and closer to the middle (0) line.

Figure 3 allows transcending the trajectory characteristics perspective and observing the resulting transition matrices, represented via connectivity plots showing the leading edges connecting buildings and their land use. The uniqueness of the autonomous w/ car scenario is clearly visible - the network is much denser, mobility is organized around an entirely different set of hubs or central

¹ Whether this result is sensible for a small case study area is debatable; under the supervised implementation, differences are minimal.

locations, and travel distances are higher. Most notably, the Mahne Yehuda Market which is the main hub under the supervised scenarios and to a lesser extent also under the autonomous w/o car implementation, has no significant role. Overall, the resulting connectivity plot looks synthetic and unaligned with common mobility patterns. This is perhaps due to the LLM forcefully trying to implement car-based logics to a small scale area. A slight impression of synthetic results also emerges for the connectivity matrix for the autonomous w/o car scenario, whose degree of hierarchical order is lower in comparison with the plots generated under the supervised scenarios. Notice that edge lengths under the supervised w/ car scenario tend to be longer than for its w/o car counterpart. This suggests that the transition matrix constraint does not stop profile-based differences from being significant, even if subtle.

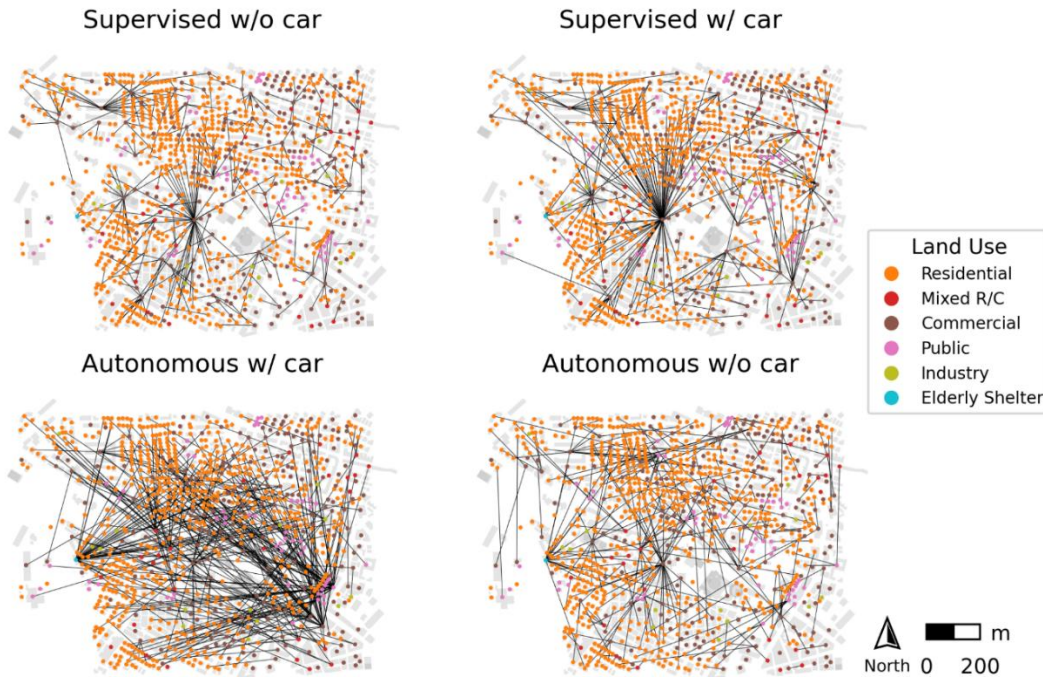


Figure 3. Connectivity plots by implementation scenario. Only top 0.03% edges are presented to avoid clutter.

5. Conclusion

This paper explores two issues in using LLMs to generate synthetic trajectories: producing aggregate transition matrices sensitive to socio-demographic attributes, and assessing the impact of autonomy on outcomes. As an initial exploration, we focus on distribution shifts rather than real-world validation.

Our results show meaningful variation across autonomy and profiles. This finding is notable given that the profile difference reduces to a single word in the prompt (“You have **a** car” vs. “You have **no** car”). The supervised mode naturally reproduces gravity-based patterns even though the choice set is unweighted. This suggests that the LLM internalizes distance-decay and building-size preferences regardless of explicit probability weights and supports plausibility. However, the gravity-based transition matrix generates pressure to return home, evident in hallucinations emerging almost exclusively under this regime and in the dominance of residential visits from the

third stop onwards. The LLM tendency to accommodate car-based mobility dynamics is most visible in the car-based autonomous scenario, producing a connectivity plot entirely different to the gravity-aligned no-car version. This indicates that the model can generate transport-mode-adjusted distributions without additional empirical information. Yet results appear synthetic under the autonomous mode, indicating that behavioral constraints are still needed.

An intermediate solution is thus required, one that combines the best of both worlds. This may require more structured guidance for the search procedure, better specification of spatial relations (e.g. explicitly using coordinates, as some of the documented reasoning procedures did try to access this unavailable information), or allowing the model to violate transition rules through a controlled process. These directions, along with validation in relation to real world data, mark the primary and immediate avenues for further research.

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Appendix

A.1 Prompts for Behavioral Context

Supervised: “Land use encoding - 1: residential, 2: mixed residential and commercial, 3: commercial, 4: public, 5: industry, 6: elderly shelter. You should generate a trajectory with arbitrary length in this conversation, where each response you must choose a building to visit next from the provided building list, the adjacent buildings will then be provided from user again until reaching home again. The trajectory should always start from and return to home. Note that, building count is not a proxy of time, as it is possible to stay at a building for long hours, but you should include these reasoning. We assume that no building will be visited twice except for home. For each iteration, output time, uniq_id, behavioral reasoning in json format. You are an employee aged {age}. You have {a / no} car. Your home is {home_id}.”

Autonomous: “Land use encoding - 1: residential, 2: mixed residential and commercial, 3: commercial, 4: public, 5: industry, 6: elderly shelter. You should generate a trajectory with arbitrary length with first-person behavioral reasoning based on these buildings. The trajectory should always start from and return to home. Note that, building count is not a proxy of time, as it is possible to stay at a building for long hours, but you should include these reasoning. We assume that no building will be visited twice except for home. Output time, uniq_id, behavioral reasoning in json format. You are an employee aged {age}. You have {a / no} car. Your home is {home_id}.”

A.2 Connectivity Plot: Simple Gravity Model

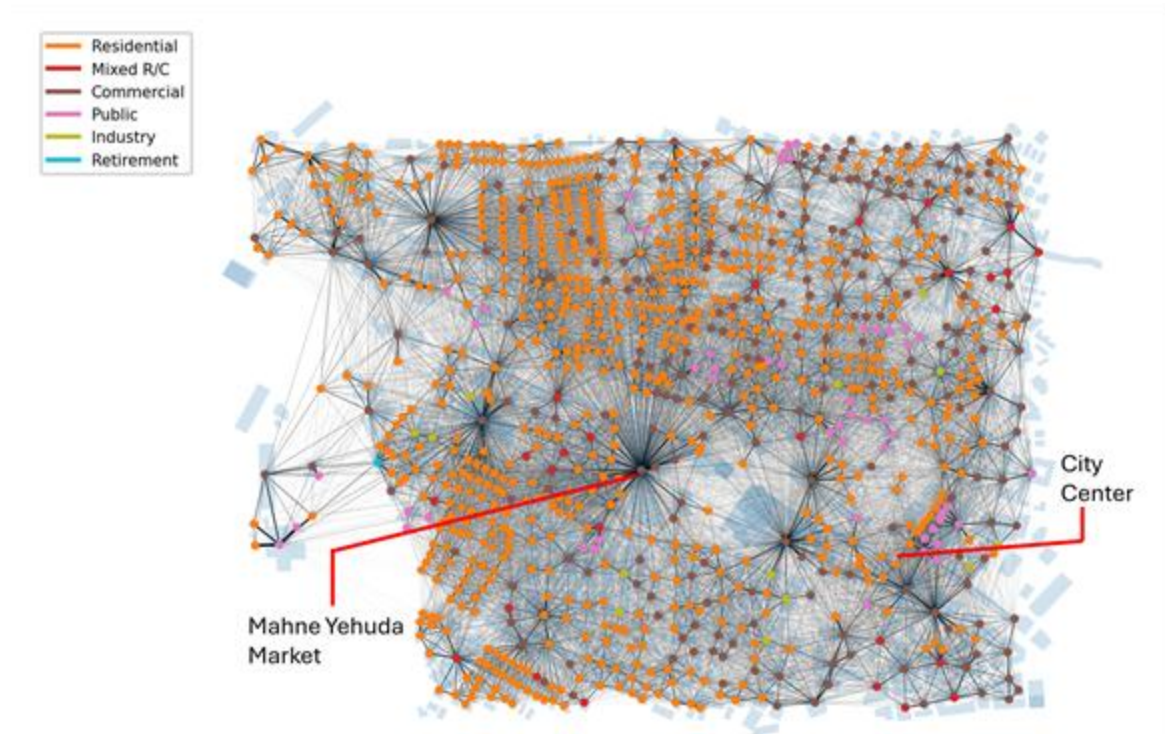


Figure A.2. Connectivity plot based on a simple gravity model.